What's Good About TensorFlow 2.0?

Version 2.0 of TensorFlow is focused on simplicity and ease of use. It has been strengthened with updates like eager execution and intuitive higher level APIs accompanied by flexible model building. It is platform agnostic, and makes APIs more consistent, while removing those that are redundant.

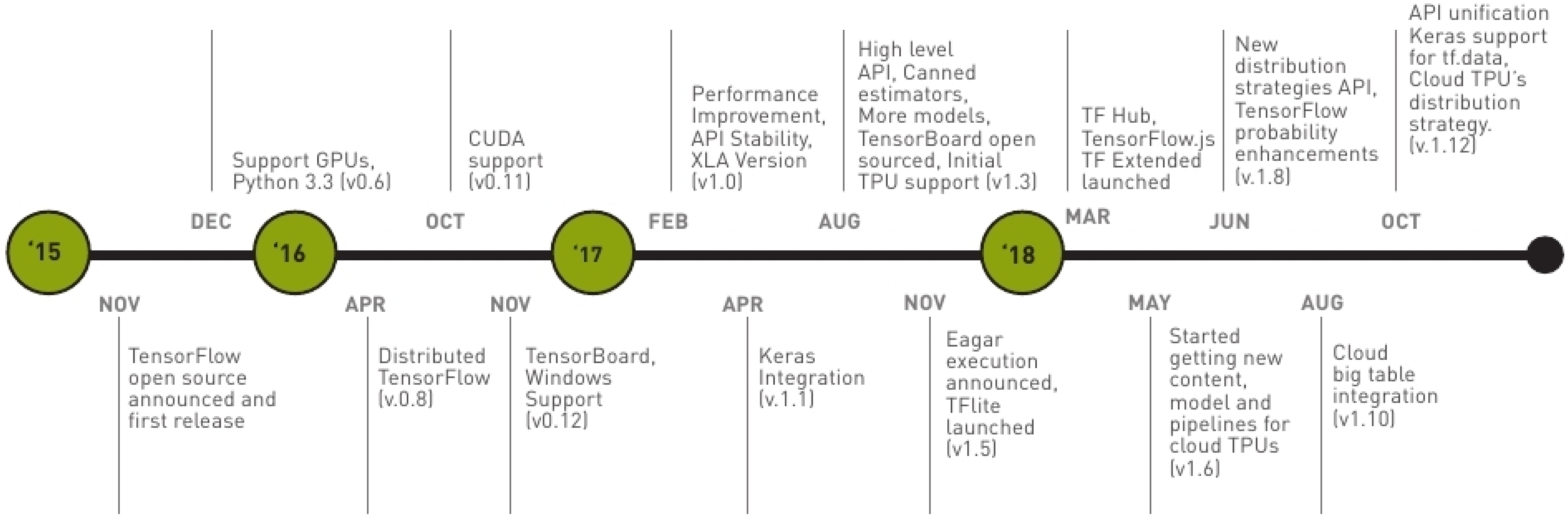

Figure 1: The evolution of TensorFlow (Image source: TensorFlow Developer Summit 2019)

Machine learning and artificial intelligence are experiencing a revolution these days, primarily due to three major factors. The first is the increased computing power available within small form factors such as GPUs, NPUs and TPUs. The second is the breakthrough in machine learning algorithms. State-of-art algorithms and hence models are available to infer faster. Finally, huge amounts of labelled data is essential for deep learning models to perform better, and this is now available.

TensorFlow is an open source AI framework from Google which arms researchers and developers with the right tools to build novel models. It was made open source in 2015 and, in the past few years, has evolved with various enhancements covering operator support, programming languages, hardware support, data sets, official models, and distributed training and deployment strategies.

TensorFlow 2.0 was released recently at the TensorFlow Developer Summit. It has major changes across the stack, some of which will be discussed from the developers' point of view.

TensorFlow 2.0 is primarily focused on the ease-of-use, power and scalability aspects. Ease is ensured in terms of simplified APIs, Keras being the main high level API interface; eager execution is available by default. Version 2.0 is powerful in the sense of being flexible and running much faster than earlier, with more optimisation. Finally, it is more scalable since it can be deployed on high-end distributed environments as well as on small edge devices.

This new release streamlines the various components involved, from data preparation all the way up to deployment on various targets. High speed data processing pipelines are offered by *tf.data*, high level APIs are offered by *tf.keras*, and there are simplified APIs to access various distribution strategies on targets like the CPU, GPU and TPU. TensorFlow 2.0 offers a unique packaging format called SavedModel that can be deployed over the cloud through a TensorFlow Serving. Edge devices can be deployed through TensorFlow Lite, and Web applications through the newly introduced TensorFlow.js and various other language bindings that are also available.

TensorFlow.js was announced at the developer summit with off-the-shelf pretrained models for the browser, node, desktop and mobile native applications. The inclusion of Swift was also announced. Looking at some of the performance improvements since last year, the latest release claims a training speedup of 1.8x on NVIDIA Tesla V100, a 1.6x training speedup on Google Cloud TPUv2 and a 3.3.x inference speedup on Intel Skylake.

Upgrade to 2.0

The new release offers a utility *tf_upgrade_v2* to convert a 1.x Python application script to a 2.0 compatible script. It does most of the job in converting the 1.x deprecated